

Python Foundations

Interactive I/O, Arithmetic, and Real-World Logic

Input & Type Casting

Analyzing `demo0-en.py`

Handling User Data

By default, `input()` captures all keystrokes as a String. To perform mathematical operations like age calculation, we must cast the input string into an integer using `int()`.

```
# Reading text
name = input("What is your name? ")

# Reading an integer (requires casting)
birth_year = int(input("What year were you born? "))

# Processing
est_age = 2026 - birth_year

# Output
print("Hello,", name, "!")
print("You are approx.", est_age, "years old.")
```



Arithmetic Operators

Analyzing `demo1-en.py`

Beyond Basic Calculators

Standard Operations

Python supports all foundational math logic natively. Note that standard division (/) always produces a Float.

```
a = 15
b = 4

print("Add:", a + b)    # 19
print("Sub:", a - b)    # 11
print("Mult:", a * b)   # 60
print("Div:", a / b)    # 3.75
```

Algorithmic Operators

These special operators are heavily utilized in advanced Computer Science logic, encryption, and loops.

```
# Special algorithm operators
print("Int Div:", a // b) # 3 (Quotient)
print("Modulo:", a % b)   # 3 (Remainder)
print("Power:", a ** b)   # 50625
```

Understanding Modulo (%)

3

The Result of $15 \% 4$

The modulo operator does not divide. Instead, it returns the **remainder** of a division operation.

- ▶ The number 4 fits into 15 exactly 3 times ($4 \times 3 = 12$).
- ▶ The difference between 15 and 12 is 3.
- ▶ Modulo is primarily used to determine if a number is even/odd ($n \% 2 = 0$) or to bound numbers within arrays.

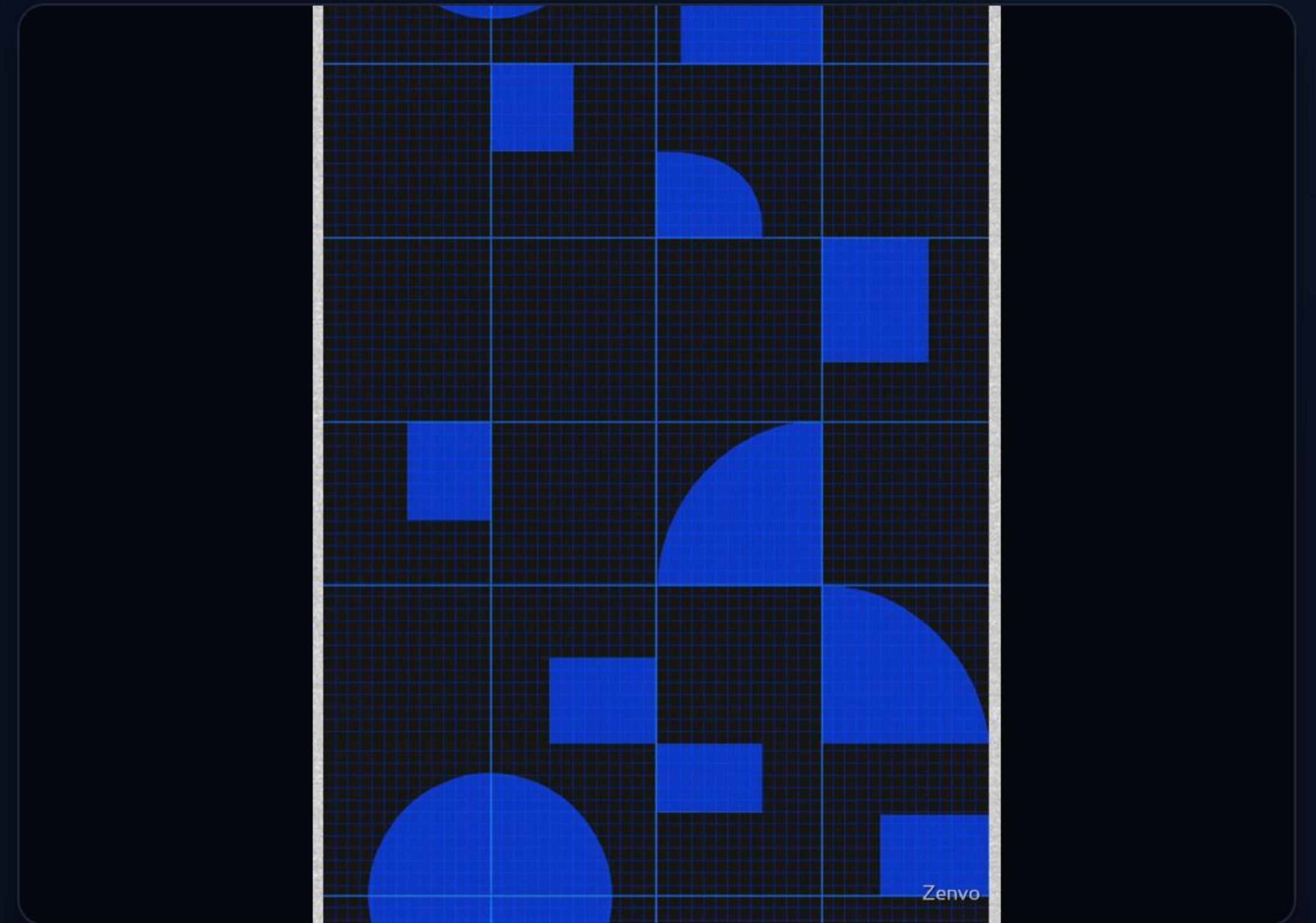
Practical Geometry

Analyzing `demo2-en.py`

Area and Perimeter Logic

By assigning inputs to descriptive variables (L for length, l for width), we construct clear geometric formulas in the Processing stage.

```
# Problema: Se citesc lungimea ...  
# 1. DATE DE INTRARE  
L = int(input("Introdu lungimea ... : "))  
l = int(input("Introdu lățimea ... : "))  
  
# 2. PRELUCRARE (Calculule)  
aria = L * l  
perimetrul = 2 * (L + l)  
  
# 3. DATE DE IEȘIRE  
print("Aria dreptunghiului este:", aria)  
print("Perimetrul ... este:", perimetrul)
```



The Taxi Fare Problem

We are tasked with building a dynamic financial calculator based on user variables (`demo3-en.py`).

A taxi charges a flat **base fare** plus a variable **rate per mile**.

Given the final **total cost** paid by a client, our program must reverse-engineer the math to output exactly how many miles they traveled.



Isolating Variables in Code

Mathematical Approach

To solve for miles, we must isolate the distance cost first by subtracting the base fare from the total, then divide by the per-mile rate.

- ▶ We cast inputs to `float()` because currency implies decimals.
- ▶ `dist_cost = total - base`
- ▶ `miles = dist_cost / rate`

Implementation

```
# 1. INPUT
base = float(input("Base fare ($): "))
rate = float(input("Rate per mile ($): "))
total = float(input("Total cost ($): "))

# 2. PROCESSING
dist_cost = total - base
miles = dist_cost / rate

# 3. OUTPUT
print("The client traveled:", miles, "miles.")
```

End of Demos

Questions regarding logic extraction, casting, or operators?

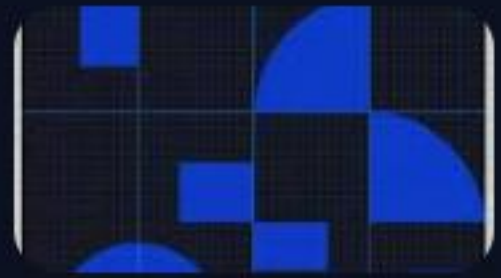
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